

What Is Soil, and What Is It For? Social Ontologies and Social Epistemologies of Soil Affordances and Soil Experiments

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ABSTRACT ORIGINAL

Soil has been defined in numerous ways within the scientific study of soil. How soil has been defined and what it is thought to be for have influenced the design of soil experiments and the development of agricultural technologies aimed at its improvement. Normative goals, like improving soil capabilities, increasing crop yield, and sustainably managing soil health for future generations structure the kinds of questions being pursued in soil science research and in turn shape what can be known from agronomy experiments. I analyse how the interplay of epistemological, normative, and ontological conceptualizations of soil have recursively shaped soil knowledge, what we think soil is for, and how soil is categorized. I focus on three case studies to do this: (1) J. H. Quastel's conceptualization of soil metabolism, (2) the development of the Haber-Bosch process and widespread use of chemical nitrogen fertilizers; and (3) George Washington Carver's long-term soil improvement experiments. These illustrate both how social epistemologies and ontologies shape development of new agricultural biotechnologies and how adoption of these affords different opportunities and burdens to farmers in virtue of how these technologies affect the chemical, physical, and biological interactions of soil-plant-animal interactions. © The Author(s) 2025.